

Shuoming Zhang (张朔铭)

Ph.D. Candidate in Computer Architecture

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Research Interests

Kernel agents (LLM-driven generation, optimization, and verification of low-level compute kernels); large-scale multi-agent systems; LLM infrastructure and serving systems, with an emphasis on constrained and structured decoding; reliability and safety for LLM-driven systems; compiler design and code translation.

Education

Ph.D. in Computer Architecture, Institute of Computing Technology, CAS & UCAS 2021 – present
Advisors: Prof. Huimin Cui and Assoc. Prof. Jiacheng Zhao. State Key Laboratory of Processors (SKLP). Beijing

B.Eng. in Computer Science, University of Chinese Academy of Sciences (UCAS) 2017 – 2021

Selected Honors & Awards

- **Distinguished Paper Award**, CGO 2026 — *From Threads to Tiles (T2T)*.
- **Spotlight**, ICML 2026 — *CONTINUUM*.
- **Hygon Ph.D. Award**, 2024.

Publications

[†] equal contribution * corresponding author. Author name in **bold**.

Peer-Reviewed

1. **[CCS 2026]** *When Grammar Guides the Attack: Uncovering Control-Plane Vulnerabilities in LLMs with Structured Output*
Shuoming Zhang, Jiacheng Zhao*, Hanyuan Dong, Ruiyuan Xu, Zhicheng Li, Yangyu Zhang, Shuaijiang Li, Yuan Wen, Chunwei Xia, Zheng Wang, Xiaobing Feng, Huimin Cui
[arXiv:2503.24191](https://arxiv.org/abs/2503.24191)
2. **[CCF THPC]** *The New Compiler Stack: A Survey on the Synergy of LLMs and Compilers*
Shuoming Zhang, Jiacheng Zhao*, Qiuchu Yu, Chunwei Xia, Zheng Wang, Xiaobing Feng, Huimin Cui
3. **[CCF THPC]** *LEGO-Compiler: Enhancing Neural Compilation Through Translation Composability*
Shuoming Zhang, Jiacheng Zhao, Chunwei Xia, Zheng Wang, Yunji Chen, Xiaobing Feng, Huimin Cui*
4. **[EMNLP 2024, findings]** *Introducing Compiler Semantics into Large Language Models as Programming Language Translators: A Case Study of C to x86 Assembly*
Shuoming Zhang, Jiacheng Zhao, Chunwei Xia, Zheng Wang, Yunji Chen, Huimin Cui*
5. **[ICML 2026, spotlight]** *CONTINUUM: Restoring the Contiguous Tensor Abstraction Efficiently for Dynamic AI Workloads via Hardware Virtualization*
Yangyu Zhang[†], **Shuoming Zhang**[†], Chunwei Xia, Shuaijiang Li, Zhicheng Li, Ruiyuan Xu, Zheming Yang, Lei Chen, Yuan Wen, Guangli Li, Xiaobing Feng, Huimin Cui, Jiacheng Zhao*
6. **[ICML 2026]** *LEGO: An LLM-Enabled Hierarchical Optimizer for Tensor Computation Graphs with Structure-Aware Search and Compositional Synthesis*
Ruiyuan Xu[†], **Shuoming Zhang**[†], Guangli Li, Qiuchu Yu, Rui Zhang, Yangyu Zhang, Hao Qian, Chunwei Xia, Jiacheng Zhao, Chenxi Wang, Xiaobing Feng, Jingling Xue, Huimin Cui
7. **[OSDI 2026]** *MoonBright: A GPU Memory Allocator with Device-Side Page Table Materialization and Deferred TLB Coherence*
Yangyu Zhang, Lei Chen, Chunwei Xia, Shuaijiang Li, **Shuoming Zhang**, Zhicheng Li, Qianqi Sun, Jiawei Xiao,

Ruiyuan Xu, Ao Chen, Guangli Li, Xiaobing Feng, Huimin Cui, Chenxi Wang, Jiacheng Zhao*
[USENIX](#)

8. **[ISCA 2026]** *Symbiotic MLLM Serving: Dynamically Balancing Parallelism Across GPUs and Resources Within GPUs*
Zhicheng Li, Jiacheng Zhao*, Yangyu Zhang, Zhaolin Duan, Xinyu Liu, Siqu Li, **Shuoming Zhang**, Shuaijiang Li, Donglin Yu, Yuan Wen, Chunwei Xia, Xiyu Shi, Huimin Cui
9. **[CGO 2026, Distinguished Paper Award]** *From Threads to Tiles: T2T, a Compiler for CUDA-to-NPU Translation via 2D Vectorization*
Shuaijiang Li, Jiacheng Zhao*, Ying Liu, **Shuoming Zhang**, Lei Chen, Yijin Li, Yangyu Zhang, Zhicheng Li, Runyu Zhou, Xiyu Shi, Chunwei Xia, Yuan Wen, Xiaobing Feng, Huimin Cui
10. **[SCIS 2026]** *Large Processor Chip Model*
Kaiyan Chang, Mingzhi Chen, Yunji Chen*, ..., Rui Zhang, **Shuoming Zhang**, Jiacheng Zhao (alphabetical order, equal contributions)
[Springer](#)
11. **[NeurIPS 2025, poster]** *SpaceServe: Spatial Multiplexing of Complementary Encoders and Decoders for Multimodal LLMs*
Zhicheng Li, **Shuoming Zhang**, Jiacheng Zhao*, Siqu Li, Xiyu Shi, Yangyu Zhang, Shuaijiang Li, Donglin Yu, Zheming Yang, Yuan Wen, Huimin Cui
[OpenReview](#)
12. **[NeurIPS 2025, poster]** *Mutual-Supervised Learning for Sequential-to-Parallel Code Translation*
Changxin Ke, Rui Zhang*, Shuo Wang, Li Ding, Guangli Li, Yuanbo Wen, **Shuoming Zhang**, Ruiyuan Xu, Jin Qin, Jiaming Guo, Chenxi Wang, Ling Li, Qi Guo, Yunji Chen
[OpenReview](#)

Preprints / Work-in-Progress

1. **[arXiv 2026]** *Decode-Time Grammars: Constrained LLM Generation over a Refinement Order of Grammar Fragments*
Shuoming Zhang, Ruiyuan Xu, Haofeng Li, Qiuchu Yu, Yangyu Zhang, Chunwei Xia, Xiaobing Feng, Chenxi Wang, Huimin Cui, Jiacheng Zhao*
[arXiv:2607.18357](#)
2. **[Architecture 2.0 @ ISCA 2026, workshop]** *From Skills to Tracelets: Dependency-Guided Transfer for LLM Kernel Agents*
Shuoming Zhang, Ruiyuan Xu, Qiuchu Yu, Guangli Li, Huimin Cui, Jiacheng Zhao*
[OpenReview](#)
3. **[arXiv 2026]** *Learning When to Optimize: Verified Optimization Skills from Expert GPU-Kernel Lineages (KLineage)*
Shuoming Zhang[†], Qiuchu Yu[†], Yangyu Zhang, Ruiyuan Xu, Xiyu Shi, Guangli Li, Xiaobing Feng, Huimin Cui, Jiacheng Zhao*
[arXiv:2605.28213](#)

Research Experience

Current projects

- **Kernel agents.** Agentic pipelines that synthesize, tune, and verify high-performance kernels for AI accelerators with LLM-in-the-loop search and feedback.
- **Large-scale multi-agent systems.** Orchestration and runtime support for populations of cooperating LLM agents on complex software-engineering and system-level tasks.
- **Constrained and structured decoding.** Decoding as LLM infrastructure: constrained and grammar-based decoding for controllable, reliable, and safe generation (e.g., structured-output control-plane analysis).

- **Grammar-guided code generation.** Decode-time grammars that steer LLMs with grammar constraints to produce syntactically and semantically valid code, eliminating invalid references by construction (arXiv 2026).

Earlier projects

- **LLM-guided compilation workflows.** Model-in-the-loop pipelines for source-to-assembly translation and error recovery with LLM feedback.
- **LLM-aware compiler construction.** Reusable compiler components leveraging LLM reasoning for IR transformation, code generation, and verification.
- **Heterogeneous model offloading with TVM** (collaboration with Intel). NPU/CPU co-execution and scheduling within the TVM stack; prototyped a new TVM backend for a simulator-based NPU.
- **VLIW instruction scheduling** (collaboration with Huawei). Instruction-scheduling heuristics targeting domain-specific VLIW architectures.

Internships

XCoreSigma — Research Intern.

2024 – present

Intel — Research Intern; TVM compiler stack for heterogeneous NPU/CPU offloading.

Feb – May 2023

Huawei — Research Intern; compiler back-end and VLIW instruction scheduling.

Feb – Jun 2021

Professional Service

Reviewer: Transactions on Machine Learning Research (TMLR), Artificial Intelligence Review, ICML, NeurIPS, ICLR, AAAI 2027.

Technical Skills

Languages: C/C++, Python, CUDA. **Systems & Tools:** LLVM/MLIR, TVM, PyTorch, LLM serving/inference stacks, Git, Linux.

Areas: compiler construction, code generation & translation, GPU/NPU kernel optimization, LLM agents and serving systems.